Exp 22 MATLAB CODE

```
clc;
clear;
close all;

% Parameters
N = 1000;
SNR = 10;

% Generate Random Binary Data
data = randi([0 1],1,N);

% BPSK Modulation
bpsk = 2*data - 1;

% Add AWGN Noise
rx = awgn(bpsk,SNR,'measured');

% Constellation Diagram
figure;

subplot(1,2,1);
plot(real(bpsk),imag(bpsk),'bo');
title('BPSK Constellation Diagram');
xlabel('In-Phase');
ylabel('Quadrature');
grid on;
axis([-2 2 -2 2]);
```

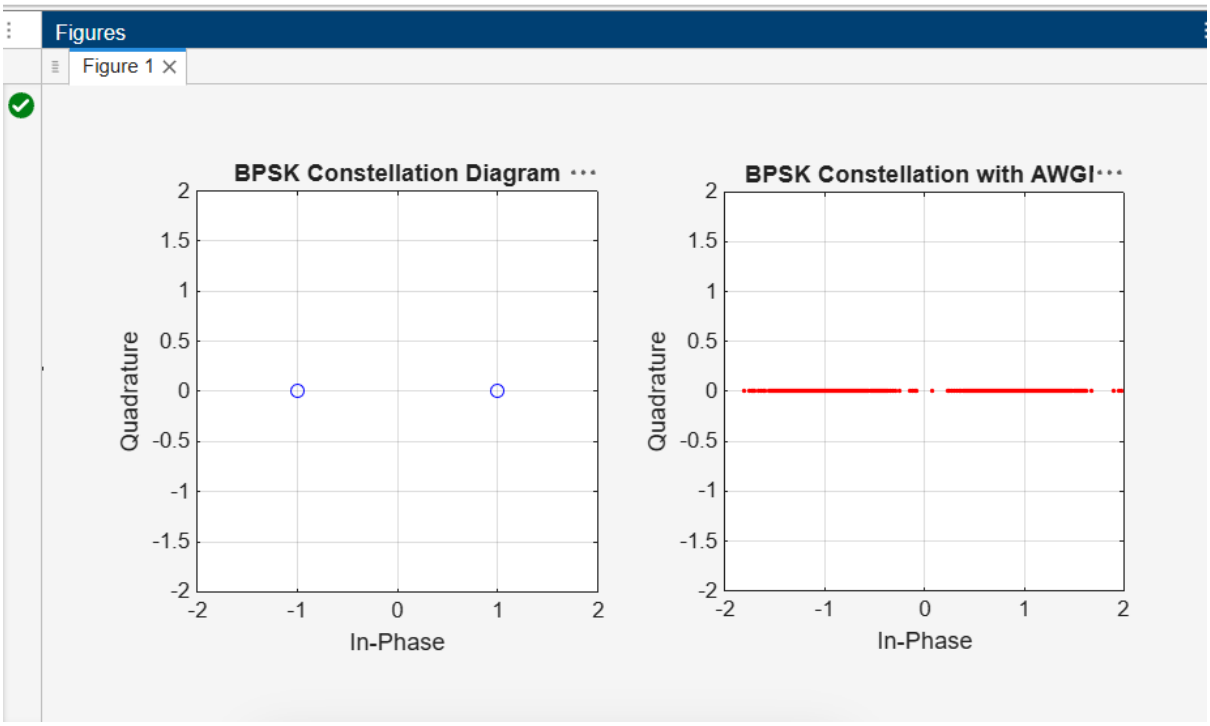
```
axis square;

subplot(1,2,2);
plot(real(rx),imag(rx),'r.');
title('BPSK Constellation with AWGN');
xlabel('In-Phase');
ylabel('Quadrature');
grid on;
axis([-2 2 -2 2]);
axis square;

% Demodulation
detected_data = rx > 0;

% Calculate BER
errors = sum(data ~= detected_data);
BER = errors/N;

% Display Results
fprintf('SNR = %d dB\n',SNR);
fprintf('Total Bits = %d\n',N);
fprintf('Bit Errors = %d\n',errors);
fprintf('Bit Error Rate (BER) = %.6f\n',BER);
```


Figures
Figure 1
BPSK Constellation Diagram
Quadrature
In-Phase
2
1.5
1
0.5
0
-0.5
-1
-1.5
-2
BPSK Constellation with AWGI
Quadrature
In-Phase